

# Applied Physics Lab

Diploma in Engineering and Technology • Semester 1 & 2 • Basic Science

## Course Code

# 1006

Base course code: 1006

Credits: 2 • L:0, T:0, P:2

Periods/Semester: 30 + 30

## Subject Type

### Physics Lab

This file adapts the lesson structure according to the subject type, so the learning material behaves like a textbook, revision guide, workbook and answer guide.

## Programme

### Diploma in Engineering and Technology

Designed for Kerala Polytechnic students with simple explanations, exam-friendly formats and practical support notes.

## Exam / Evaluation

### Evaluation Style

Minimum 12 experiments; CIA in both semesters; ESE at end of second semester.

## Student-friendly subject summary

Applied Physics Lab builds the foundation needed for the semester by connecting syllabus outcomes with real study practice. Read the overview first, then open one module at a time, revise key points, and finally practise the expected questions with the master answer key.

**Malayalam tip:** **ഓരോ module-യും open ചെയ്ത് പഠിക്കുക.** **അവസാനം മാസ്റ്റർ അൻസ്വർ കീ.**

## Prerequisite Knowledge

- Basic knowledge in Physics from secondary school.

## Course Objectives

- Supplement physics theory through first-hand manipulation of apparatus.
- Develop scientific temper for engineering and technology problems.
- Build confidence in handling equipment and measurement skills.

## How to study this subject

- Read the module outcome before studying the topic.
- Write definitions, formulas/key points and procedures in a separate revision page.
- Practise expected questions after completing each module.
- Use the Master Answer Key to improve answer order and presentation.

## Course Outcomes

| CO  | Outcome                                                       | Hours | Level    |
|-----|---------------------------------------------------------------|-------|----------|
| CO1 | Select measuring tools and make accurate measurements.        | 12    | Applying |
| CO2 | Apply mechanics and properties of matter through experiments. | 23    | Applying |
| CO3 | Use lens, prism and glass slab to realise ray optics laws.    | 6     | Applying |
| CO4 | Use V-I characteristics to determine resistance of materials. | 15    | Applying |

## Module Map

| Module   | Title                                                                | Hours | Linked outcome                                                |
|----------|----------------------------------------------------------------------|-------|---------------------------------------------------------------|
| Module 1 | Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer | 12    | Select measuring tools and make accurate measurements.        |
| Module 2 | Mechanics and Properties of Matter Experiments                       | 23    | Apply mechanics and properties of matter through experiments. |
| Module 3 | Ray Optics Experiments                                               | 6     | Use lens, prism and glass slab to realise ray optics laws.    |
| Module 4 | Electrical and Semiconductor Experiments                             | 15    | Use V-I characteristics to determine resistance of materials. |

## Quick navigation cards

Mini Formula Bank

# Mini Formula Bank for Applied Physics Lab

Use this as the last-minute memory page before exams and practical sessions.

## Least count

**Smallest measurable value of an instrument**

**Where used:** Measurement experiments

**Common mistake:** Write least count before observations.

## Pendulum relation

$$T = 2\pi\sqrt{L/g}$$

**Where used:** Simple pendulum

**Common mistake:** Length must be from point of suspension to bob centre.

## Series resistance

$$R = R_1 + R_2 + \dots$$

**Where used:** Resistance law

**Common mistake:** Series increases total resistance.

## Lens power

$$P = 1/f(m)$$

**Where used:** Optics

**Common mistake:** Focal length must be in metre.

## Hooke's law

$$F = kx$$

**Where used:** Spring experiment

**Common mistake:** Valid only within elastic limit.

## Ohm's law

$$V = IR$$

**Where used:** V-I characteristics

**Common mistake:** Temperature should remain nearly constant.

## Parallel resistance

$$1/R = 1/R_1 + 1/R_2 + \dots$$

**Where used:** Resistance law

**Common mistake:** Equivalent is less than smallest branch resistance.

Module 1 • 12 hours

# Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

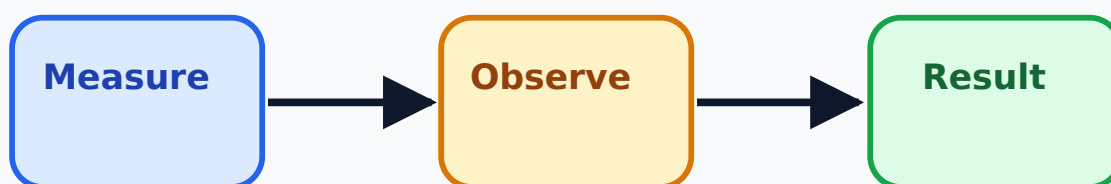
**Learning outcome:** Select measuring tools and make accurate measurements.

Lab record-□ least count, zero error, units □□□□□□ skip □□□□□□□□□□.

## Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.

### Measurement Tools: Vernier, Screw Gaug



Aim → Apparatus → Procedure → Observation → Calculation → Result

## Topic-wise explanation

### Vernier calipers volume measurement

Vernier calipers volume measurement is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Screw gauge diameter/thickness

Screw gauge diameter/thickness is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Spherometer radius of curvature

Spherometer radius of curvature is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Mercury thermometer

Mercury thermometer is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Least count

Least count is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Zero error

Zero error is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Unit conversion

Unit conversion is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Observation table

Observation table is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Important terms

| Term                                | Exam-ready meaning                                                                                                                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vernier calipers volume measurement | Vernier calipers volume measurement is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example. |
| Screw gauge diameter/thickness      | Screw gauge diameter/thickness is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.      |
| Spherometer radius of curvature     | Spherometer radius of curvature is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.     |
| Mercury thermometer                 | Mercury thermometer is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.                 |
| Least count                         | Least count is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.                         |

## Practical / real-life examples

### Example

Volume of cylinder =  $\pi r^2 h$  using vernier readings.

### Example

Correct reading = observed reading  $\pm$  zero correction.

**Exam writing tip: Start with the exact term, then add explanation, example and conclusion.**

## Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

## Module checklist

Outcome read

Definitions ready

Examples practised

Diagram/table revised

Questions attempted

Mistakes checked

## Expected Questions

1. **M1-Q1** Explain Vernier calipers volume measurement with its importance in Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.
2. **M1-Q2** Write short notes on Screw gauge diameter/thickness.
3. **M1-Q3** Prepare an exam-ready answer for Spherometer radius of curvature.
4. **M1-Q4** Compare or connect two key ideas from Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

5. **M1-Q5** Draw/describe a neat labelled diagram, table or format related to Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.
6. **M1-Q6** List common mistakes/precautions while studying or performing Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Module 2 • 23 hours

## Mechanics and Properties of Matter Experiments

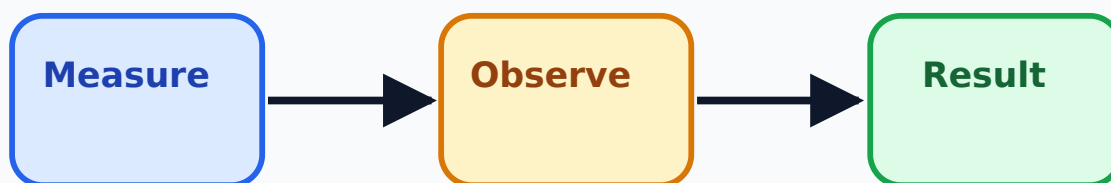
**Learning outcome:** Apply mechanics and properties of matter through experiments.

Repeated readings → mean value → result reliable.

### Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.

#### Mechanics and Properties of Matter Exp



Aim → Apparatus → Procedure → Observation → Calculation → Result

### Topic-wise explanation

#### Hooke's law and spring constant

Hooke's law and spring constant is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to

avoid. In exam answers, keep the order logical and use labelled points.

### **Stoke's law viscosity**

Stoke's law viscosity is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Parallelogram law of forces**

Parallelogram law of forces is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Moment bar**

Moment bar is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **U-tube relative density**

U-tube relative density is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Flywheel moment of inertia**

Flywheel moment of inertia is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Simple pendulum g**

Simple pendulum g is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Resonance column velocity of sound**

Resonance column velocity of sound is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### **Cantilever oscillation**

Cantilever oscillation is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.



# Important terms

| Term                            | Exam-ready meaning                                                                                                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hooke's law and spring constant | Hooke's law and spring constant is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example. |
| Stoke's law viscosity           | Stoke's law viscosity is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.           |
| Parallelogram law of forces     | Parallelogram law of forces is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.     |
| Moment bar                      | Moment bar is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.                      |
| U-tube relative density         | U-tube relative density is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.         |

# Practical / real-life examples

## Example

Hooke's law graph: load vs extension should be a straight line within elastic limit.

## Example

Simple pendulum: measure time for many oscillations to reduce error.

**Exam writing tip: Start with the exact term, then add explanation, example and conclusion.**

# Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

# Module checklist

Outcome read

Definitions ready

Examples practised

Diagram/table revised

Questions attempted

Mistakes checked

# Expected Questions

1. **M2-Q1** Explain Hooke's law and spring constant with its importance in Mechanics and Properties of Matter Experiments.
2. **M2-Q2** Write short notes on Stoke's law viscosity.
3. **M2-Q3** Prepare an exam-ready answer for Parallelogram law of forces.
4. **M2-Q4** Compare or connect two key ideas from Mechanics and Properties of Matter Experiments.
5. **M2-Q5** Draw/describe a neat labelled diagram, table or format related to Mechanics and Properties of Matter Experiments.
6. **M2-Q6** List common mistakes/precautions while studying or performing Mechanics and Properties of Matter Experiments.

**Module 3 • 6 hours**

## Ray Optics Experiments

**Learning outcome:** Use lens, prism and glass slab to realise ray optics laws.

Optics-□ pin alignment/parallax error marks loss □□□□ □□□□□□□□.

### Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.

## Ray Optics Experiments



Aim → Apparatus → Procedure → Observation → Calculation → Result

## Topic-wise explanation

### Convex lens focal length by u-v method

Convex lens focal length by u-v method is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Power of lens

Power of lens is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Snell's law

Snell's law is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Glass slab/prism refraction

Glass slab/prism refraction is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Object and image distance

Object and image distance is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Ray diagram

Ray diagram is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Refractive index

Refractive index is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Important terms

| Term                                   | Exam-ready meaning                                                                                                                                         |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Convex lens focal length by u-v method | Convex lens focal length by u-v method is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example. |
| Power of lens                          | Power of lens is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.                          |
| Snell's law                            | Snell's law is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.                            |
| Glass slab/prism refraction            | Glass slab/prism refraction is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.            |
| Object and image distance              | Object and image distance is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.              |

## Practical / real-life examples

### Example

Lens formula:  $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$  (sign convention as taught).

### Example

Power  $P = \frac{1}{f}$  in metre, unit dioptre.

**Exam writing tip: Start with the exact term, then add explanation, example and conclusion.**

## Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

## Module checklist

Outcome read

Definitions ready

Examples practised

Diagram/table revised

Questions attempted

Mistakes checked

## Expected Questions

1. **M3-Q1** Explain Convex lens focal length by u-v method with its importance in Ray Optics Experiments.
2. **M3-Q2** Write short notes on Power of lens.
3. **M3-Q3** Prepare an exam-ready answer for Snell's law.
4. **M3-Q4** Compare or connect two key ideas from Ray Optics Experiments.
5. **M3-Q5** Draw/describe a neat labelled diagram, table or format related to Ray Optics Experiments.
6. **M3-Q6** List common mistakes/precautions while studying or performing Ray Optics Experiments.

Module 4 • 15 hours

## Electrical and Semiconductor Experiments

**Learning outcome:** Use V-I characteristics to determine resistance of materials.

Circuit ON ☐ connection faculty/technician ☐ verify ☐.

## Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.

## Electrical and Semiconductor Experiments



Aim → Apparatus → Procedure → Observation → Calculation → Result

## Topic-wise explanation

### Ohm's law V-I graph

Ohm's law V-I graph is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Resistance in series and parallel

Resistance in series and parallel is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Semiconductor diode V-I characteristics

Semiconductor diode V-I characteristics is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Knee voltage

Knee voltage is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Potentiometer internal resistance/emf comparison

Potentiometer internal resistance/emf comparison is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

### Meter bridge specific resistance

Meter bridge specific resistance is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Circuit connections

Circuit connections is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Graph plotting

Graph plotting is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

## Important terms

| Term                                             | Exam-ready meaning                                                                                                                                                                     |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ohm's law V-I graph                              | Ohm's law V-I graph is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.                              |
| Resistance in series and parallel                | Resistance in series and parallel is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.                |
| Semiconductor diode V-I characteristics          | Semiconductor diode V-I characteristics is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.          |
| Knee voltage                                     | Knee voltage is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.                                     |
| Potentiometer internal resistance/emf comparison | Potentiometer internal resistance/emf comparison is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example. |

## Practical / real-life examples

### Example

Ohm's law: slope of V-I graph gives resistance.

### Example

Meter bridge balance point is used to calculate unknown resistance.

**Exam writing tip: Start with the exact term, then add explanation, example and conclusion.**

## Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

## Module checklist

Outcome read

Definitions ready

Examples practised

Diagram/table revised

Questions attempted

Mistakes checked

## Expected Questions

1. **M4-Q1** Explain Ohm's law V-I graph with its importance in Electrical and Semiconductor Experiments.
2. **M4-Q2** Write short notes on Resistance in series and parallel.
3. **M4-Q3** Prepare an exam-ready answer for Semiconductor diode V-I characteristics.
4. **M4-Q4** Compare or connect two key ideas from Electrical and Semiconductor Experiments.
5. **M4-Q5** Draw/describe a neat labelled diagram, table or format related to Electrical and Semiconductor Experiments.
6. **M4-Q6** List common mistakes/precautions while studying or performing Electrical and Semiconductor Experiments.

### Quick Revision

## One-page revision summary

Use this section on the previous day of exam or practical assessment.

## Module-wise must-know points

- **Module 1:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer — revise Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer



radius of curvature, Mercury thermometer.

- **Module 2:** Mechanics and Properties of Matter Experiments — revise Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar.
- **Module 3:** Ray Optics Experiments — revise Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction.
- **Module 4:** Electrical and Semiconductor Experiments — revise Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage.

## Important definitions / formulas / key points

- **Least count:** Smallest measurable value of an instrument — Measurement experiments
- **Hooke's law:**  $F = kx$  — Spring experiment
- **Pendulum relation:**  $T = 2\pi\sqrt{l/g}$  — Simple pendulum
- **Ohm's law:**  $V = IR$  — V-I characteristics
- **Series resistance:**  $R = R_1 + R_2 + \dots$  — Resistance law
- **Parallel resistance:**  $1/R = 1/R_1 + 1/R_2 + \dots$  — Resistance law
- **Lens power:**  $P = 1/f(m)$  — Optics

## Common exam mistakes

- Writing general answers instead of syllabus-topic answers.
- Forgetting diagrams, tables, units, procedures or examples.
- Skipping final conclusion/result in long answers.
- Not numbering answers neatly.

## Last-minute checklist

Course outcomes

Module map

Key bank

Expected questions

Answer format

Diagrams/tables

Final revision

## Diagram / table list

Prepare neat labelled diagrams, flowcharts, observation tables, comparison tables or answer formats wherever the module asks for drawing, procedure, calculation or communication structure.

**Master Answer Key**

## Answers matching every expected question

Each answer number exactly follows the expected question number.

## **M1-Q1 — Explain Vernier calipers volume measurement with its importance in Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M1-Q2 — Write short notes on Screw gauge diameter/thickness.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M1-Q3 — Prepare an exam-ready answer for Spherometer radius of curvature.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M1-Q4 — Compare or connect two key ideas from Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M1-Q5 — Draw/describe a neat labelled diagram, table or format related to Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M1-Q6 — List common mistakes/precautions while studying or performing Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end

with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

**Module link:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

## **M2-Q1 — Explain Hooke's law and spring constant with its importance in Mechanics and Properties of Matter Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M2-Q2 — Write short notes on Stoke's law viscosity.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M2-Q3 — Prepare an exam-ready answer for Parallelogram law of forces.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M2-Q4 — Compare or connect two key ideas from Mechanics and Properties of Matter Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M2-Q5 — Draw/describe a neat labelled diagram, table or format related to Mechanics and Properties of Matter Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M2-Q6 — List common mistakes/precautions while studying or performing Mechanics and Properties of Matter Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

**Module link:** Mechanics and Properties of Matter Experiments

## **M3-Q1 — Explain Convex lens focal length by u-v method with its importance in Ray Optics Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

## **M3-Q2 — Write short notes on Power of lens.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

## **M3-Q3 — Prepare an exam-ready answer for Snell's law.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

## **M3-Q4 — Compare or connect two key ideas from Ray Optics Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

## **M3-Q5 — Draw/describe a neat labelled diagram, table or format related to Ray Optics Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end

with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

### **M3-Q6 — List common mistakes/precautions while studying or performing Ray Optics Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

**Module link:** Ray Optics Experiments

### **M4-Q1 — Explain Ohm's law V-I graph with its importance in Electrical and Semiconductor Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments

### **M4-Q2 — Write short notes on Resistance in series and parallel.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments

### **M4-Q3 — Prepare an exam-ready answer for Semiconductor diode V-I characteristics.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments

### **M4-Q4 — Compare or connect two key ideas from Electrical and Semiconductor Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel,

Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments

#### **M4-Q5 — Draw/describe a neat labelled diagram, table or format related to Electrical and Semiconductor Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments

#### **M4-Q6 — List common mistakes/precautions while studying or performing Electrical and Semiconductor Experiments.**

**Answer format:** Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

**Module link:** Electrical and Semiconductor Experiments